Engineering - Undergraduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The coil of a moving coil meter has 100 turns, is 40 mm long and 30 mm wide. The control torque is 240*10-6 N-m on full scale. If magnetic flux density is 1 Wb/m 2 range of meter is

   (a) 2 mA.

   (b) 5 mA.

   (c) 1.5 mA.

   (d) 0.5 mA.

   (e) 6 mA.

2. A sinusoidal signal is analog signal, because

   (a) it is a square wave signal

   (b) it has a constant frequency

   (c) it is positive for one half cycle

   (d) it can only have a single value at a time

   (e) it has a sinusoidal waveform

3. What is the name of the fluorescent material that gives red colour fluorescence?

   (a) Zinc silicate.

   (b) Calcium sulphide.

   (c) Zinc oxide.

   (d) Zinc sulphide.

   (e) Calcium silicate.

4. Find the Laplace transform L{g(t)}, where g(t)= 0,t $<$ 4 and= (t - 4)^2,t $\geq$ 4

(2 e^-4s)/s^3

   (a) 2 / s^3

   (b) (2 / s^2) * e^-4s

   (b) 4 / s^3

(2 e^-4s)/s^4

5. Given a lossless medium withϵ_r= 10 and μ_r = 5 find its impedance.

   (a) 132 ohms

   (b) 377 ohms

(c) 150 ohms
(d) 300 ohms
(e) 266 ohms

True / False (10 marks)

Answer True or False

6. True or False: The lowest natural frequency of the automobile engine spring is 25.8 cycles per second.
7. True or False: The Laplace transform of f(t) = t^2 is equal to 1 / s^2.
8. True or False: The one cycle surge current rating of an SCR with a half cycle surge current of 3000 A at 50 Hz is approximately 2121.32 A.
9. True or False: The capacitance of a parallel plate capacitor is incorrectly given by C = (S/d^2), which suggests area is inversely related to capacitance.
10. True or False: A spring rate of 1300 lb./in. would effectively reduce the engine's unbalanced force by 15%.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the isothermal compression of a liquid in a quasistatic process, deriving the expression for the total work done based on the relationship In(V/V) = -A(p - p).
12. Discuss the relationship between a time-varying magnetic field, described by B = B_0 $e^b ta$_z, and the corresponding induced electric field. Derive the expression for the electric field produced by this magnetic field and analyze its significance in engineering applications.

End of Paper